

Experiment:- 6

Problem statement:-

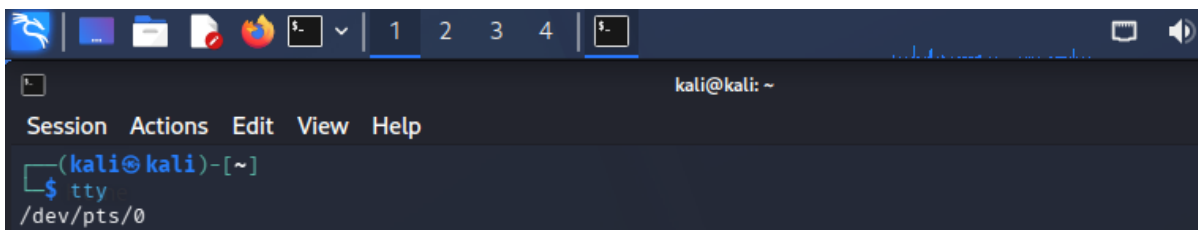
Linux operating systems support multiple user sessions and interfaces through the use of virtual terminals and graphical environments. Understanding how Linux manages text based terminals, graphical display servers, display managers, X clients, and window managers is essential for system administration, troubleshooting, and cyber security operations.

The problem is to explore and understand virtual terminals, display managers, X clients, and window managers, and to practice switching between text-based and graphical interfaces in a Linux environment.

Aim:- Understanding virtual terminal, display manager, X clients, and window manager in linux.

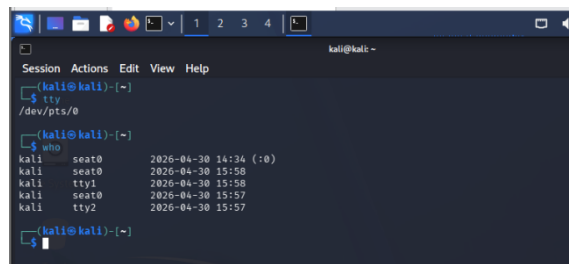
Procedure:-

1. Log in to the Linux system and open the terminal. Log in to the Linux system. Open the terminal. Check the current terminal: tty



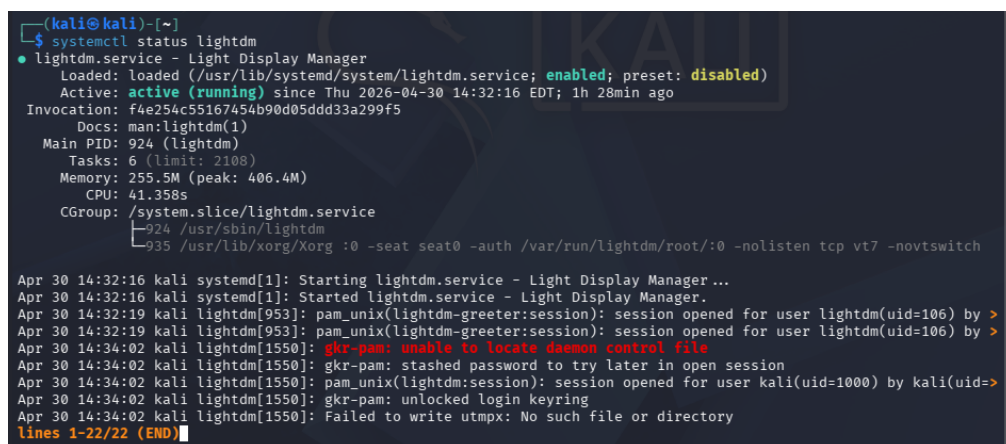
2. Switch to another virtual terminal using: Ctrl + Alt + F2. Log in and verify terminal access. Switch back to graphical mode: Ctrl + Alt + F1 or Ctrl + Alt + F7

3. Check logged-in users: who w



```
kali@kali:~$ who
kali    seat0    2026-04-30 14:34 (:0)
kali    seat0    2026-04-30 15:58
kali    tty1     2026-04-30 15:58
kali    seat0    2026-04-30 15:57
kali    tty2     2026-04-30 15:57
```

4. Check the display manager status: systemctl status lightdm



```
(kali@kali)-[~]
$ systemctl status lightdm
● lightdm.service - Light Display Manager
   Loaded: loaded (/usr/lib/systemd/system/lightdm.service; enabled; preset: disabled)
   Active: active (running) since Thu 2026-04-30 14:32:16 EDT; 1h 28min ago
     Invocation: f4e254c55167454b90d05ddd33a299f5
       Docs: man:lightdm(1)
    Main PID: 924 (lightdm)
      Tasks: 6 (limit: 2108)
     Memory: 255.5M (peak: 406.4M)
        CPU: 41.358s
    CGroup: /system.slice/lightdm.service
            └─924 /usr/sbin/lightdm
              └─935 /usr/lib/xorg/Xorg :0 -seat seat0 -auth /var/run/lightdm/root/:0 -nolisten tcp vt7 -novtswitch

Apr 30 14:32:16 kali systemd[1]: Starting lightdm.service - Light Display Manager ...
Apr 30 14:32:16 kali systemd[1]: Started lightdm.service - Light Display Manager.
Apr 30 14:32:19 kali lightdm[953]: pam_unix(lightdm-greeter:session): session opened for user lightdm(uid=106) by >
Apr 30 14:32:19 kali lightdm[953]: pam_unix(lightdm-greeter:session): session opened for user lightdm(uid=106) by >
Apr 30 14:34:02 kali lightdm[1550]: gkr-pam: unable to locate daemon control file
Apr 30 14:34:02 kali lightdm[1550]: gkr-pam: stashed password to try later in open session
Apr 30 14:34:02 kali lightdm[1550]: pam_unix(lightdm:session): session opened for user kali(uid=1000) by kali(uid=>
Apr 30 14:34:02 kali lightdm[1550]: gkr-pam: unlocked login keyring
Apr 30 14:34:02 kali lightdm[1550]: Failed to write utmpx: No such file or directory
lines 1-22/22 (END)
```

5. Verify X session variable: echo \$DISPLAY



```
(kali@kali)-[~]
$ echo $DISPLAY
:0.0

(kali@kali)-[~]
$
```

6. List running X-related processes: `ps -e | grep X`

```
(srishti_m@kali) - [~/grep_lab1]
$ ps -e | grep x
 19 ?      00:00:00 rcu_exp_par_gp_kthread_worker/0
 20 ?      00:00:00 rcu_exp_gp_kthread_worker
307 ?      00:00:00 irq/18-vmwgfx
361 ?      00:00:00 kworker/R-xt4-rsv-conversion
648 ?      00:00:00 kworker/R-xprtiod
713 ?      00:00:00 VBoxService
1048 ?     00:00:00 mpris-proxy
1064 ?     00:00:00 xfce4-session
1130 ?     00:00:00 VBoxClient
1131 ?     00:00:00 VBoxClient
1145 ?     00:00:00 VBoxClient
1147 ?     00:00:01 VBoxClient
1153 ?     00:00:00 VBoxClient
1154 ?     00:00:08 VBoxClient
1225 ?     00:00:00 VBoxClient
1226 ?     00:00:00 VBoxClient
1232 ?     00:00:10 xfwm4
1262 ?     00:00:01 xfsettingsd
1268 ?     00:00:01 xfce4-panel
1282 ?     00:00:02 xfdesktop
1333 ?     00:00:00 xfce4-notifyd
1359 ?     00:00:01 xfce4-screensav
1365 ?     00:00:00 xiccd
1376 ?     00:00:00 xfce4-power-man
1397 ?     00:00:00 xcape
```

Expected Output

- Successful switching between virtual terminals.
- Identification of display manager and X session.
- Display of active user sessions.
- Confirmation of graphical environment variables.

Result

Virtual terminals, display managers, X clients, and window managers were successfully explored. The student gained practical understanding of Linux system interfaces and graphical session management.